

Weierstrass Convergence Theorem.

Def. A sequence $f_n: G \rightarrow \mathbb{C}$ converges locally uniformly to f if:
for every compact $K \subset G$,

$$\sup_{z \in K} |f_n(z) - f(z)| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thm. Suppose that $f_n: G \rightarrow \mathbb{C}$ is holomorphic on G and $f_n \rightarrow f$ locally uniformly, then f is holomorphic. Moreover, $f'_n \rightarrow f'$ locally uniformly on G , (G open).

Proof. Given $z_0 \in G$, take $\delta > 0$ such that $\overline{B_\delta(z_0)} \subset G$. By the hypothesis, $f_n \rightarrow f$ uniformly on $\overline{B_\delta(z_0)}$, and so f is continuous on $\overline{B_\delta(z_0)}$.

Let T be a triangular path inside $\overline{B_\delta(z_0)}$. Since $f_n \rightarrow f$ uniformly,

$$\int_T f_n(z) dz \rightarrow \int_T f(z) dz. \text{ But, by the Cauchy-Goursat's theorem, each } n \in \mathbb{N}, \int_T f_n(z) dz = 0, \text{ so } \int_T f(z) dz = 0.$$

Now, by the Morera's theorem, f is Holomorphic on $\overline{B_\delta(z_0)}$.

Since z_0 was arbitrary, so f is Holomorphic on G . To show the second assertion, let $K \subset G$ be a compact subset. Since $\mathbb{C}$ is T_4 space and G is open, there is an open U such that $K \subset U \subset \overline{U} \subset G$.

Now, for each $\delta > 0$, define

$$U_\delta := \{z \in U \mid \overline{D_\delta(z)} \subset U\} \text{ where } D_\delta = \{z' \in \mathbb{C} \mid |z' - z| < \delta\}.$$

Given $z \in U_\delta$, by the Cauchy's integral formula,

$$f'(z) - f'_n(z) = \frac{1}{2\pi i} \cdot \int_{C_\delta} \frac{f(u) - f_n(u)}{(u - z)^2} du$$

where C_δ is a curve centered at z with radius $\delta > 0$. Therefore,

$$\begin{aligned} |f'(z) - f'_n(z)| &\leq \frac{1}{2\pi} \cdot \int_{C_\delta} \frac{|f(u) - f_n(u)|}{|u - z|^2} |du| \leq \frac{1}{2\pi} \cdot \frac{2\pi\delta}{\delta^2} \cdot \sup_{u \in C_\delta} |f(u) - f_n(u)| \\ &\leq \frac{1}{\delta} \cdot \sup_{u \in \overline{U}} |f(u) - f_n(u)| \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Since the bound that tends to zero is independent of choice of z ,
Therefore, we can choose $\delta > 0$ small enough to $K \subset U_\delta$,
so $f'_n \rightarrow f'$ uniformly. $\square$